

| Functional Test Cases (Postman/API) |                                                 |                                                                                                                                                                                                                                                          |                                                                                                                    |                                                                                                                                        |                                                                                           |          |
|-------------------------------------|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|----------|
| Test Case ID                        | Scenario                                        | DATA                                                                                                                                                                                                                                                     | PRECONDITION                                                                                                       | Input/ Steps                                                                                                                           | Expected Output                                                                           | SEVERITY |
| TC_F_01                             | Admin sees "Change Discount" button             | <a href="https://qwallity-dev.onrender.com">https://qwallity-dev.onrender.com</a> Login: qwallitymanual19022025@gmail.com Password: qwallity19022025 Test should be done for this Application Admin username: "Lilyalma", Password: ""Lilyalma1"         | Postman is Open, New collection is created, New Environment for variables is created. Admin user is registered     | 1. Login as admin<br>2. Navigate to User_Action page                                                                                   | "Change Discount" button visible                                                          | High     |
| TC_F_02                             | Non-admin does not see "Change Discount" button | <a href="https://qwallity-dev.onrender.com">https://qwallity-dev.onrender.com</a> Login: qwallitymanual19022025@gmail.com Password: qwallity19022025 Test should be done for this Application Non-Admin username: "Lilyalma_li", Password: "hello_lilit" | Postman is Open, New collection is created, New Environment for variables is created. Non-admin user is registered | 1. Login as non-admin<br>2. Go to User_Action                                                                                          | "Change Discount" button not visible                                                      | High     |
| TC_F_03                             | Admin updates discount                          | <a href="https://qwallity-dev.onrender.com">https://qwallity-dev.onrender.com</a> Login: qwallitymanual19022025@gmail.com Password: qwallity19022025 Test should be done for this Application Admin username: "Lilith_i", Password: "Exam111"            | Postman is Open, New collection is created, New Environment for variables is created. Admin is registered          | 1. Login as admin<br>2. Select "Fundamental" from dropdown<br>3. Enter discount = 50 and course_count =4<br>4. Click "Change Discount" | Notification: "Discount changes"                                                          | Critical |
| TC_F_04                             | Non-admin views discounted course               | <a href="https://qwallity-dev.onrender.com">https://qwallity-dev.onrender.com</a> Login: qwallitymanual19022025@gmail.com Password: qwallity19022025 Test should be done for this Application                                                            | Postman is Open, New collection is created, New Environment for variables is created. Non-admin is logged in       | 1. Navigate to course page<br>2. View Fundamental course                                                                               | Discounted price shown, notification: "You have 10% discount. Discounted price displays." | Critical |
| TC_F_05                             | Discount applied to Advanced course             | <a href="https://qwallity-dev.onrender.com">https://qwallity-dev.onrender.com</a> Login: qwallitymanual19022025@gmail.com Password: qwallity19022025 Test should be done for this Application Non-Admin username: "Lilyalma_li", Password: "hello_lilit" | Postman is Open, New collection is created, New Environment for variables is created                               | Login as user and buy Advanced course                                                                                                  | Discounted price visible                                                                  | High     |
| TC_F_06                             | Edge case - Invalid discount (non-numeric)      | <a href="https://qwallity-dev.onrender.com">https://qwallity-dev.onrender.com</a> Login: qwallitymanual19022025@gmail.com Password: qwallity19022025 Test should be done for this Application Admin username: "Lilith_i", Password: "Exam111"            | Postman is Open, New collection is created, New Environment for variables is created                               | Discount = "abc"                                                                                                                       | Validation error                                                                          | High     |
| TC_F_07                             | Edge case - Missing course_count                | <a href="https://qwallity-dev.onrender.com">https://qwallity-dev.onrender.com</a> Login: qwallitymanual19022025@gmail.com Password: qwallity19022025 Test should be done for this Application Admin username: "Lilith_i", Password: "Exam111"            | Postman is Open, New collection is created, New Environment for variables is created                               | No course_count entered                                                                                                                | Validation error                                                                          | High     |
|                                     |                                                 |                                                                                                                                                                                                                                                          |                                                                                                                    |                                                                                                                                        |                                                                                           |          |
| Performance Testing (JMeter)        |                                                 |                                                                                                                                                                                                                                                          |                                                                                                                    |                                                                                                                                        |                                                                                           |          |
| Test Case ID                        | Scenario                                        | DATA                                                                                                                                                                                                                                                     | PRECONDITION                                                                                                       | STEPS                                                                                                                                  | Expected Result                                                                           | SEVERITY |
| TC_P_01                             | Load test for "Change Discount" API             | <a href="https://qwallity-dev.onrender.com">https://qwallity-dev.onrender.com</a> Login: qwallitymanual19022025@gmail.com Password: qwallity19022025 Test should be done for this Application                                                            | JMETER is open.                                                                                                    | Simulate 100 concurrent admin requests to update discount                                                                              | Response time < 2s, success rate > 95%                                                    | Critical |
| TC_P_02                             | Load test for course purchase                   | <a href="https://qwallity-dev.onrender.com">https://qwallity-dev.onrender.com</a> Login: qwallitymanual19022025@gmail.com Password: qwallity19022025 Test should be done for this Application                                                            | JMETER is open.                                                                                                    | 200 concurrent users buying courses                                                                                                    | Response time < 3s                                                                        | Critical |
| TC_P_03                             | Stress test for discount update                 | <a href="https://qwallity-dev.onrender.com">https://qwallity-dev.onrender.com</a> Login: qwallitymanual19022025@gmail.com Password: qwallity19022025 Test should be done for this Application                                                            | JMETER is open.                                                                                                    | Increase requests until system fails                                                                                                   | System handles at least 500 requests/min                                                  | High     |
| TC_P_04                             | Spike test for user purchase flow               | <a href="https://qwallity-dev.onrender.com">https://qwallity-dev.onrender.com</a> Login: qwallitymanual19022025@gmail.com Password: qwallity19022025 Test should be done for this Application                                                            | JMETER is open.                                                                                                    | Sudden spike from 10 to 100 users                                                                                                      | System stabilizes quickly                                                                 | High     |
| TC_P_05                             | Endurance test                                  | <a href="https://qwallity-dev.onrender.com">https://qwallity-dev.onrender.com</a> Login: qwallitymanual19022025@gmail.com Password: qwallity19022025 Test should be done for this Application                                                            | JMETER is open.                                                                                                    | Simulate users buying courses over 2 hrs                                                                                               | No memory leaks, consistent response time                                                 | High     |
|                                     |                                                 |                                                                                                                                                                                                                                                          |                                                                                                                    |                                                                                                                                        |                                                                                           |          |
| MySQL Workbench DB Test Cases       |                                                 |                                                                                                                                                                                                                                                          |                                                                                                                    |                                                                                                                                        |                                                                                           |          |
| Test Case ID                        | Scenario                                        | DATA                                                                                                                                                                                                                                                     | Precondition                                                                                                       | Query                                                                                                                                  | Expected Output                                                                           | SEVERITY |

## CMD Test Cases

| Test Case ID | Scenario                          | DATA                                                                                                                                                                                          | PRECONDITION                      | Command                                                                                                                 | Expected Output                 | SEVERITY |
|--------------|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------|---------------------------------|----------|
| TC_CMD_01    | API call to fetch discount config | <a href="https://qwallity-dev.onrender.com">https://qwallity-dev.onrender.com</a> Login: qwallitymanual19022025@gmail.com Password: qwallity19022025 Test should be done for this Application | Wind + R open CMD (ready to work) | curl -X GET <a href="http://localhost/api/discounts">http://localhost/api/discounts</a>                                 | JSON with discount info         | Medium   |
| TC_CMD_02    | Admin updates discount via CLI    | <a href="https://qwallity-dev.onrender.com">https://qwallity-dev.onrender.com</a> Login: qwallitymanual19022025@gmail.com Password: qwallity19022025 Test should be done for this Application | Wind + R open CMD (ready to work) | curl -X POST -d '{ "course_type": "Advanced", "discount": 50, "course_count": 4}' http://localhost/api/discounts/update | Response: "Discount changes"    | Critical |
| TC_CMD_03    | User fetches course price         | <a href="https://qwallity-dev.onrender.com">https://qwallity-dev.onrender.com</a> Login: qwallitymanual19022025@gmail.com Password: qwallity19022025 Test should be done for this Application | Wind + R open CMD (ready to work) | curl http://localhost/api/courses                                                                                       | Prices displayed with discount  | High     |
| TC_CMD_04    | Authenticated user fetch          | <a href="https://qwallity-dev.onrender.com">https://qwallity-dev.onrender.com</a> Login: qwallitymanual19022025@gmail.com Password: qwallity19022025 Test should be done for this Application | Wind + R open CMD (ready to work) | Using curl with token header                                                                                            | Only authorized user can access | High     |
| TC_CMD_05    | Test invalid input                | <a href="https://qwallity-dev.onrender.com">https://qwallity-dev.onrender.com</a> Login: qwallitymanual19022025@gmail.com Password: qwallity19022025 Test should be done for this Application | Wind + R open CMD (ready to work) | Post with missing field                                                                                                 | Returns error response          | Medium   |